Preservation of Causal Versus Correlational Language During LLM-Assisted Editing

Abstract

Large language models (LLMs) are increasingly used to revise, condense, clarify, and polish scholarly manuscripts. Although such assistance can improve fluency and readability, editing can also alter the epistemic force of scientific statements. One particularly consequential transformation is the substitution of correlational language with causal language—or the reverse. This distinction matters because causal claims assert or imply that changing one variable would alter another, whereas correlational claims describe statistical or empirical association without necessarily identifying a causal mechanism. Causal inference frameworks have long emphasized that the evidentiary requirements for causal claims differ from those for associational claims, particularly in observational research (Rubin, 1974; Pearl, 2009). Existing research shows that scientific articles frequently contain inconsistent causal language and that inappropriate causal phrasing can influence how readers interpret the evidentiary strength of research findings. More recently, NLP systems have been developed to detect causal language in scientific writing, while studies of LLMs demonstrate weaknesses in recognizing causal distinctions, especially when claims are hedged or context-dependent. At the same time, empirical evidence indicates that LLMs are becoming embedded in scientific writing workflows. This paper synthesizes these literatures and conceptualizes causal-language preservation as a constrained editing problem: an editor should improve form, clarity, and coherence while preserving the original claim type, evidential qualification, directionality, scope, and uncertainty. The paper reviews current approaches, including causal-language classifiers, prompt-based safeguards, human-in-the-loop editing, claim verification, structured representations, and causal reasoning benchmarks. It argues that existing LLM-editing evaluation is insufficient because conventional measures of grammaticality, similarity, or factuality do not adequately capture changes in epistemic modality and causal force. Future research should develop causality-aware editing benchmarks, claim-level provenance representations, counterfactual semantic tests, domain-specific evaluation, and interactive systems that explicitly flag causal-strength changes rather than silently rewriting them.

Keywords: large language models; LLM-assisted editing; causal language; correlational language; causal inference; scientific writing; semantic preservation; epistemic modality; claim verification; NLP

1. Introduction

Large language models have rapidly become tools for drafting and revising academic prose. Their use ranges from copyediting and paraphrasing to restructuring arguments, shortening manuscripts, improving readability, and generating alternative formulations. Evidence from biomedical publishing suggests that LLM-assisted writing has already produced measurable changes in scholarly language: analysis of more than 15 million PubMed abstracts estimated that at least 13.5% of 2024 abstracts showed textual signals associated with LLM processing, although this estimate represents a lower bound rather than direct identification of individual LLM use (Kobak et al., 2025).

The increased use of LLMs creates a distinctive problem for scientific editing. A conventional copyeditor is generally expected to preserve the author's substantive meaning while improving expression. An LLM, however, does not operate solely as a mechanical transformation system. It generates new sequences according to learned linguistic and contextual distributions. Consequently, a superficially appropriate paraphrase can change the scientific proposition being expressed.

The distinction between *association* and *causation* is particularly vulnerable. Consider:

> “Higher exposure was associated with lower mortality.”
>
> and
>
> “Higher exposure reduced mortality.”

The second statement is not merely a stylistic variant of the first. “Associated with” reports an empirical relationship; “reduced” conventionally communicates a causal effect. The difference may therefore affect what readers infer about study design, confounding, intervention, and evidential strength. Causal-inference theory explicitly distinguishes statistical association from causal effects and provides formal frameworks for determining when causal interpretations are justified (Rubin, 1974; Pearl, 2009).

This issue is not hypothetical. Research on scientific publications has documented inconsistent movement between causal and non-causal formulations within the same article. Olarte Parra et al. (2021), for example, found that among observational studies published in *The BMJ*, 20% used causal language inconsistently, frequently because different sections of the paper communicated different levels of causal interpretation. Related work has found widespread use of causal terminology in observational research and systematic reviews, even when the underlying research intent does not consistently support such language (Han & Guyatt, 2020; Olarte Parra et al., 2022).

The problem is amplified by LLMs because they can perform precisely the type of lexical substitutions that encode causal strength. A model may replace “was associated with” by “was linked to,” “predicted,” “influenced,” “increased,” or “led to,” depending on context. Some of these expressions are ambiguous; others clearly imply causality. Conversely, an LLM may weaken a legitimate causal statement by replacing “caused” with “was associated with,” thereby reducing the specificity of a scientifically supported conclusion.

This paper examines the preservation of causal versus correlational language during LLM-assisted editing. It addresses five questions:

1. What makes causal-language preservation a distinct semantic problem?
2. What evidence exists concerning human and computational inconsistencies in causal language?
3. What do current LLM studies reveal about causal-language understanding?
4. Which editing and verification approaches can preserve causal force?
5. What research infrastructure is needed to evaluate and control causal-semantic drift?

The central argument is that LLM-assisted scientific editing should be evaluated not merely for grammatical quality or factuality, but for **claim-type preservation**: an edited sentence should retain the original relationship between evidence and assertion unless the user explicitly authorizes substantive revision.

2. Background and Related Work

2.1 Causal versus correlational meaning

Causal inference concerns questions about what would happen under intervention or counterfactual change, whereas statistical association describes patterns in observed data. Rubin's potential-outcomes framework formalized causal effects in terms of contrasts between potential outcomes under different treatments (Rubin, 1974). Pearl's structural causal-model framework subsequently provided graphical and structural tools for distinguishing observational association from interventional and counterfactual quantities (Pearl, 2009).

The distinction does not imply that observational data can *never* support causal inference. Modern causal inference explicitly permits causal estimation from nonrandomized data under appropriate assumptions, designs, and identification strategies. Thus, an editing system should not implement the simplistic rule “observational study → no causal language.” Contemporary methodological discussions have emphasized that causal interpretation can be justified in some observational settings when assumptions and identification strategies are appropriately specified.

This point creates an important requirement for LLM editing: **causal preservation must be evidence-sensitive rather than based solely on study-design keywords**.

2.2 Inconsistency in scientific causal language

Scientific authors themselves do not always maintain consistent causal framing. Olarte Parra et al. (2021) examined observational studies in *The BMJ* and found that causal language frequently differed between objectives, methods, results, and conclusions. In their sample, 48% of eligible articles were classified as consistently causal, 32% as consistently non-causal, and 20% as inconsistent.

A systematic survey by Olarte Parra et al. (2022) similarly found that causal intent and causal language did not reliably coincide in systematic reviews of observational studies. Among reviews with causal intent, only 4.4% used causal language consistently throughout the manuscript; among reviews without causal intent, 51.2% nevertheless used causal language somewhere in the text.

These findings establish an important baseline: causal-semantic instability is not created by LLMs. Human scientific writing already contains it. LLM editing introduces an additional transformation layer through which existing ambiguity can either be preserved, amplified, or suppressed.

Table 1 summarizes the empirical findings on causal-language use and LLM-associated writing reported in the studies discussed in Sections 1, 2.2, and 2.3.

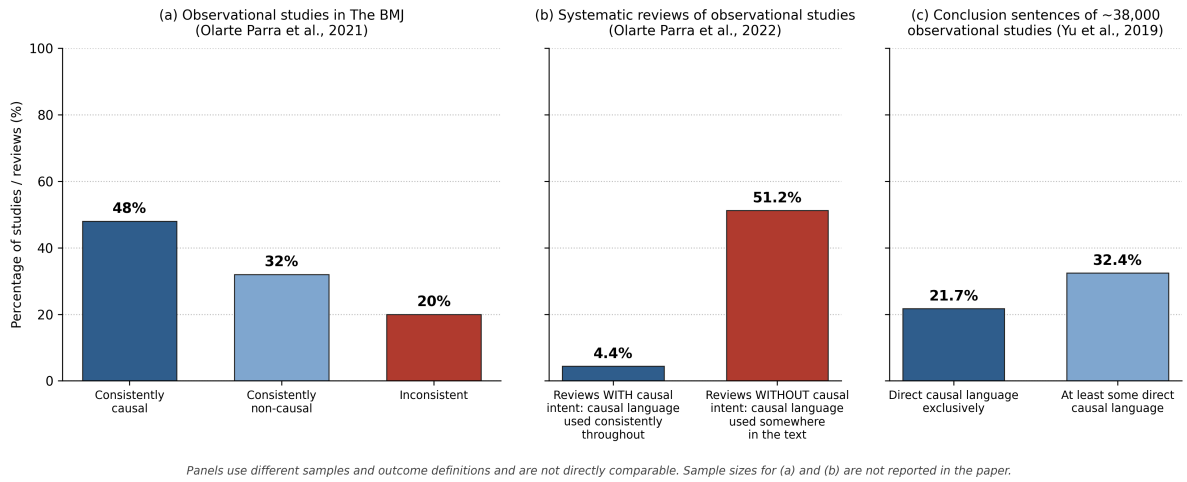
Table 1. Empirical evidence on causal-language use and LLM-associated writing in the literature. Summary of published findings on causal-language consistency in scientific reports and on the prevalence of LLM-associated vocabulary in biomedical abstracts. Samples and outcome definitions differ across studies.

Study	Data	Outcome	Reported result
Olarte Parra et al. (2021)	Observational studies in The BMJ (sample size not reported in the paper)	Consistency of causal language across objectives, methods, results, conclusions	48% consistently causal; 32% consistently non-causal; 20% inconsistent
Olarte Parra et al. (2022)	Systematic reviews of observational studies (sample size not reported)	Agreement between causal intent and causal language	Reviews with causal intent: 4.4% used causal language consistently throughout. Reviews without causal intent: 51.2% used causal language somewhere
Yu, Li & Wang (2019), applied model	~38,000 observational studies	Causal language in conclusions	21.7% used direct causal language exclusively; 32.4% used at least some
Kobak et al. (2025)	>15 million PubMed abstracts	Textual signals associated with LLM processing	At least 13.5% of 2024 abstracts (a lower bound, not identification of individual LLM use)

2.3 Computational detection of causal language

NLP research has approached the problem as classification and information extraction. Yu, Li, and Wang (2019) constructed a corpus of more than 3,000 PubMed research-conclusion sentences and classified them into “no relationship,” “correlational,” “conditional causal,” and “direct causal” categories. Their BERT-based model achieved 0.90 accuracy and a macro-F1 of 0.88. Applying the model to approximately 38,000 observational studies, they found that 21.7% used direct causal language exclusively in their conclusions and 32.4% used at least some direct causal language.

Figure 1 presents the percentages reported by Olarte Parra et al. (2021, 2022) and Yu et al. (2019).



Panels use different samples and outcome definitions and are not directly comparable. Sample sizes for (a) and (b) are not reported in the paper.

Figure 1. Reported prevalence and inconsistency of causal language in prior studies. (a) Share of observational studies in The BMJ that were consistently causal, consistently non-causal, or inconsistent (Olarte Parra et al., 2021). (b) Systematic reviews of observational studies: percentage with causal intent that used causal language consistently, and percentage without causal intent that used causal language somewhere (Olarte Parra et al., 2022). (c) Direct causal language in the conclusions of approximately 38,000 observational studies (Yu et al., 2019). Panels use different samples and definitions and are not directly comparable.

This work is highly relevant to LLM-assisted editing because it demonstrates that causal strength can be operationalized as a textual classification problem. However, classification alone does not solve preservation. A system can identify the original sentence as correlational but still produce a causal rewrite.

Causal-language detection therefore needs to be integrated into a **source–edit comparison framework**:

Original claim type → LLM transformation → Edited claim type.

The desired transformation for ordinary copyediting is generally:

$$C_{\text{original}} = C_{\text{edited}}'$$

where *C* denotes causal status, while changes in grammar, vocabulary, sentence structure, and readability remain permissible.

2.4 LLMs and causal reasoning

Evidence increasingly suggests that LLMs cannot be assumed to reliably distinguish semantic relatedness from causal explanation. Kim et al. (2023) evaluated ChatGPT on scientific and news claims categorized as causal, conditional causal, correlational, or unrelated. ChatGPT performed substantially worse than existing fine-tuned BERT models and had particular difficulty with conditional causal claims involving hedges.

More recent work provides a broader picture. Saklad et al. (2026) evaluated LLM causal inference on real-world academic literature and reported substantial difficulty with causal relationships in complex texts; the best-performing model achieved an average F1 of 0.535 on their benchmark. Liu, Baraniuk, and Sonkar (2026), using the CLEAR-3K benchmark, similarly found that LLMs frequently confuse semantic similarity with causal explanatory

relationships. Feng et al. (2025) further investigated LLM reliability for causal discovery and found limitations relevant to the use of LLMs as causal reasoning systems rather than merely language generators.

Together, these studies suggest that fluency should not be treated as evidence that an LLM has preserved causal semantics.

Table 2 summarizes the computational studies discussed in Sections 2.3 and 2.4.

Table 2. Computational studies of causal-language detection and LLM causal reasoning. Prior work evaluating classifiers and LLMs on causal-language and causal-relationship tasks. Metrics are not comparable across studies. “NR” means not reported in the paper.

Study	Task	Data / model	Reported findings
Yu et al. (2019)	4-class causal-language classification (no relationship, correlational, conditional causal, direct causal)	>3,000 PubMed conclusion sentences; BERT-based	Accuracy 0.90; macro-F1 0.88
Kim et al. (2023)	Classifying causal, conditional causal, correlational, or unrelated claims	Scientific and news claims; ChatGPT vs. fine-tuned BERT	ChatGPT substantially worse than BERT; particular difficulty with hedged conditional causal claims; results vary with prompting. Numeric values NR
Saklad et al. (2026)	LLM causal inference on real-world academic text	Benchmark of academic literature; models NR	Best model average F1 = 0.535
Liu, Baraniuk & Sonkar (2026)	Causal explanatory capability	CLEAR-3K	LLMs often confuse semantic similarity with causal explanation. Numeric values NR
Feng et al. (2025)	Reliability of LLMs for causal discovery	NR	Limitations reported. Specifics NR

3. Main Technical Discussion: Causal-Semantic Preservation as an Editing Constraint

3.1 Editing versus rewriting

LLM-assisted editing should be distinguished from unrestricted generation. In a writing task, the user's implicit objective is often:

$$\max_y Q(y)$$

subject to

$$M(y, x) = 1,$$

where x is the source text, y is the edited text, Q measures qualities such as fluency and clarity, and M represents semantic preservation.

For scientific editing, however, semantic preservation is multidimensional. At minimum, the system should preserve:

- proposition content;

- direction of association or effect;
- population and scope;
- temporal relationship;
- study-design interpretation;
- uncertainty and hedging;
- modality;
- causal versus correlational status;
- strength of the claim;
- attribution of the claim;
- conditions under which the claim holds.

Thus:

$$M = (M_p, M_d, M_s, M_t, M_h, M_c, M_m, M_a).$$

The causal component M_c is particularly important because lexical changes can modify causal interpretation without changing the principal nouns or numerical values.

3.2 A causal-strength continuum

Causal language is not binary. Scientific prose can express several levels of epistemic and causal commitment (Table 3):

Table 3. Levels of causal and epistemic commitment in scientific prose. Levels of causal commitment with example sentences and typical interpretations. Boundaries are context-dependent.

Level	Example	Typical interpretation
No relationship	“No association was observed.”	No detected relationship
Correlational	“X was associated with Y.”	Statistical association
Predictive	“X predicted Y.”	Predictive relationship; not necessarily causal
Conditional causal	“X may contribute to Y.”	Limited or qualified causal interpretation
Strong causal	“X caused Y.”	Explicit causal claim
Intervention-oriented	“Increasing X reduced Y.”	Causal intervention/effect

The boundaries are context-dependent. “Predicted,” for example, may be purely statistical or may be interpreted causally depending on the domain and surrounding text. Likewise, “risk factor” can carry causal implications even though it is frequently used ambiguously. Reporting guidelines therefore emphasize that causal implications can be conveyed through terms such as “effect,” “reduce,” “increase,” and “impact,” rather than only through the word “cause.”

Figure 2 presents the levels listed in Table 3 graphically.

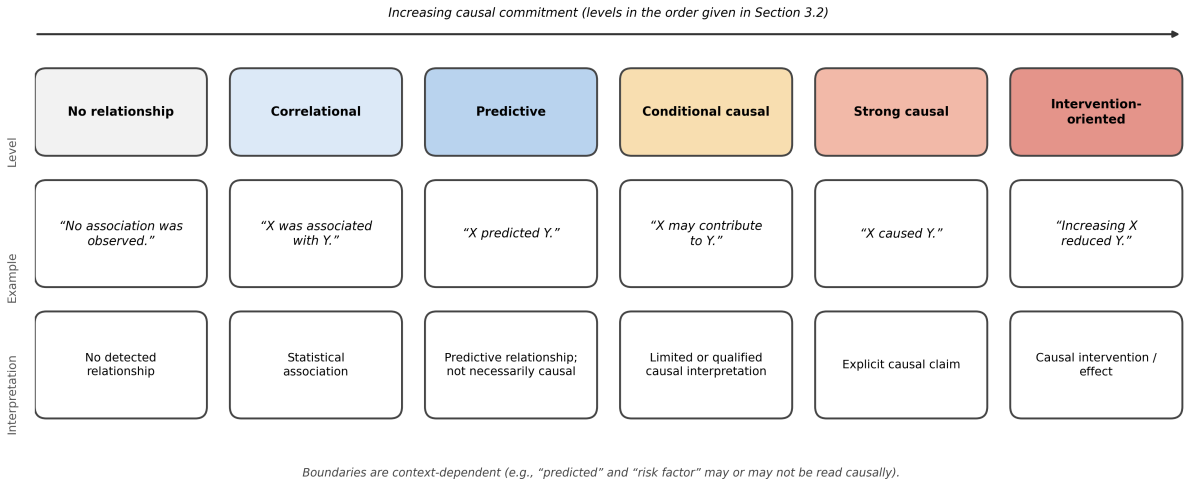


Figure 2. Levels of causal commitment in scientific language. Six levels with example sentences and typical interpretations, ordered as in Section 3.2. Boundaries are context-dependent; "predicted" and "risk factor" may or may not be read causally.

A robust editor should consequently model **causal strength as a structured property**, rather than searching for individual causal keywords.

3.3 Causal drift during paraphrasing

Causal drift occurs when an editing operation changes the relationship between evidence and claim. Four common patterns can be distinguished.

Causal inflation occurs when an associational statement becomes causal:

> Original: "Exercise was associated with lower mortality."
>
> Edited: "Exercise reduced mortality."

Causal attenuation occurs when a justified causal statement becomes merely associational:

> Original: "Randomization showed that the intervention reduced mortality."
>
> Edited: "The intervention was associated with lower mortality."

Hedge deletion occurs when uncertainty disappears:

> Original: "The findings suggest that X may influence Y."
>
> Edited: "X influences Y."

Scope expansion occurs when a qualified causal statement becomes general:

> Original: "Among adults with condition A, treatment X reduced symptom severity."
>
> Edited: "Treatment X reduces symptoms."

Table 4 summarizes the four patterns of causal drift described above.

Table 4. Patterns of causal drift during LLM paraphrasing. Four patterns by which an edit changes the relationship between evidence and claim.

Pattern	Definition	Original	Edited
Causal inflation	Associational statement becomes causal	“Exercise was associated with lower mortality.”	“Exercise reduced mortality.”
Causal attenuation	Justified causal statement becomes merely associational	“Randomization showed that the intervention reduced mortality.”	“The intervention was associated with lower mortality.”
Hedge deletion	Uncertainty disappears	“The findings suggest that X may influence Y.”	“X influences Y.”
Scope expansion	Qualified causal statement becomes general	“Among adults with condition A, treatment X reduced symptom severity.”	“Treatment X reduces symptoms.”

These transformations are difficult to detect using conventional similarity metrics because the sentences may remain lexically and syntactically close.

3.4 Why factuality is insufficient

Traditional factuality evaluation asks whether an output is supported by a source. However, causal drift demonstrates that an output can contain all the same entities and approximate factual content while changing the *type of inference*.

For example, an original study may report an association between A and B. An edited sentence stating that A “increased” B may contain no false entity, date, or numerical value. Nevertheless, it asserts something stronger than the source establishes.

Consequently, LLM editing requires at least two separate dimensions:

$$\text{Fidelity} = \text{Propositional Fidelity} + \text{Epistemic/Causal Fidelity}.$$

The latter cannot be reduced to conventional factual consistency.

4. Current Approaches and Developments

4.1 Causal-language classification

The most direct approach is to classify source and edited sentences according to causal strength. The taxonomy developed by Yu et al. (2019)—nonrelationship, correlational, conditional causal, and direct causal—provides a useful foundation.

An editing system could classify both versions:

$$P(C \mid x),\ P(C \mid y)$$

and calculate a causal-drift measure:

$$D_C(x, y) = \mathbb{1}[C_x \neq C_y].$$

A more sophisticated metric could penalize changes according to their direction and magnitude. Causal inflation should generally receive a different penalty from causal

attenuation because the epistemic consequences differ.

Figure 3 illustrates the source–edit comparison framework introduced in Section 2.3, together with the drift measure defined above.

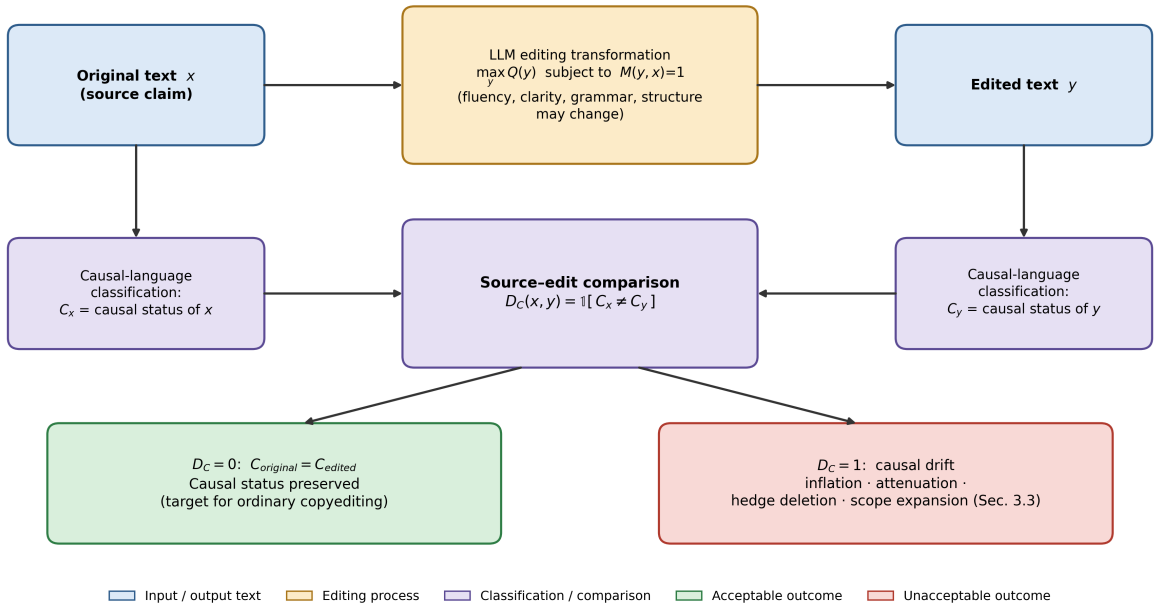


Figure 3. Conceptual framework for causal-status preservation in LLM-assisted editing. The source text *x* is edited to *y* under a semantic-preservation constraint. Causal status is classified for both versions (C_x, C_y), and drift is $D_C = 1[C_x \neq C_y]$. Ordinary copyediting should yield $C_{\text{original}} = C_{\text{edited}}$.

4.2 Prompt-based preservation

A simple operational safeguard is to instruct the LLM explicitly:

> "Improve grammar and clarity without changing whether any statement is causal, correlational, predictive, conditional, or uncertain. Preserve all hedges and causal qualifiers."

This strategy is attractive because it requires no model modification. However, Kim et al. (2023) demonstrate that prompt-based performance on causal-language tasks can vary with prompting conditions, and they caution against assuming that prompt engineering produces stable improvements across tasks and models.

Prompting should therefore be treated as a useful control layer rather than a guarantee.

4.3 Claim-level intermediate representations

A stronger architecture would first extract a structured representation of each scientific claim:

⟨*Subject*, *Predicate*, *Object*, *RelationType*, *Qualifier*, *Population*, *Time*, *Evidence*⟩.

For example:

⟨*Exercise*, associated-with, *Mortality*, observational, *Adults*, 5 years, *Cohort*⟩.

The LLM could then edit the prose while a separate verifier checks whether the structured claim remains unchanged.

This approach resembles the broader movement toward structured causal representations. Scientific claim-verification research has increasingly argued that causal structure should be explicitly represented rather than relying only on surface-level textual similarity (e.g., work on qualitative causal structures in scientific claims).

4.4 Human-in-the-loop verification

Human review remains particularly important when editing changes verbs that carry causal implications. Rather than silently applying a transformation, an editing system could present:

> **Potential causal-strength change:**
> "was associated with" → "reduced"
>
> This edit may strengthen the claim from correlational to causal.

The user can then accept, reject, or inspect the supporting evidence.

Such interfaces would shift LLM editing from autonomous rewriting toward **decision-support editing**, preserving authorial responsibility.

Figure 4 illustrates the claim-preserving editing pipeline described in Sections 4.3 and 4.4, with provenance mapping (Section 6.4) shown as a future direction.

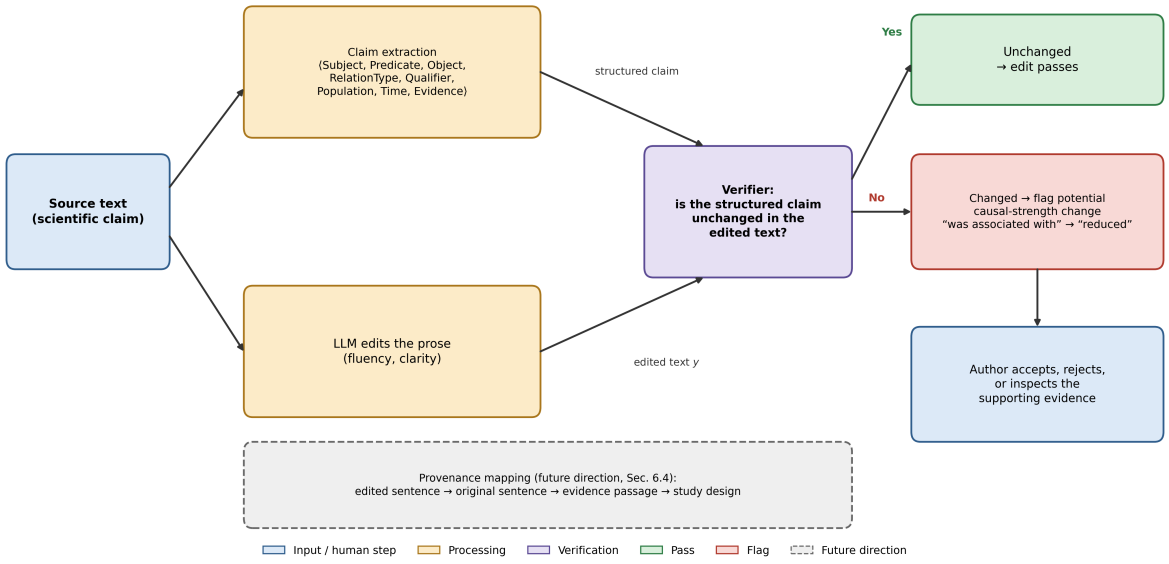


Figure 4. Proposed claim-preserving editing pipeline. A structured claim is extracted from the source. A verifier checks whether it is unchanged in the LLM-edited text. Changes are flagged for author review rather than silently applied. Provenance mapping (dashed) is a future direction. This architecture is conceptual and not evaluated in the paper.

4.5 Claim verification and evidence alignment

Causal preservation can also be connected to claim verification. A sentence asserting that an intervention “reduced” an outcome should be checked against the study design, statistical analysis, and stated conclusions.

However, verification should not automatically infer causality from study design alone. Randomized trials can still have limitations, and observational analyses can sometimes support causal inference under explicit assumptions. Current reporting guidance increasingly emphasizes transparent causal assumptions rather than an absolute experimental/observational dichotomy.

4.6 Knowledge editing and causal representations

A parallel literature on editing the internal knowledge of LLMs is also relevant. Knowledge-editing research seeks to modify specific model behaviors or facts while minimizing unintended effects on unrelated knowledge (Yao et al., 2023). Recent work has explicitly explored causal tracing and causally guided knowledge editing, illustrating growing interest in causal structure within model modification itself.

This literature is conceptually related but distinct from the present problem. Knowledge editing changes the model; causal-language preservation concerns changes in **user-provided text**. Nevertheless, techniques for tracing internal causal pathways may eventually contribute to more controllable editing systems.

Table 5 summarizes the approaches discussed in Sections 4.1 to 4.6.

Table 5. Approaches to preserving causal force in LLM-assisted editing. Approaches discussed in the paper, with the role and caveat the paper attributes to each. None is empirically evaluated in the paper.

Approach	Mechanism	Role or strength	Caveat stated in paper
Causal-language classification (4.1)	Classify source and edit; compute $D_C = 1[C_x \neq C_y]$	Operationalizes causal strength; can weight inflation and attenuation differently	Classification alone does not stop a causal rewrite (Sec. 2.3)
Prompt-based preservation (4.2)	Instruct the LLM to preserve claim type and hedges	No model modification needed	Performance varies with prompting; a control layer, not a guarantee
Claim-level representations (4.3)	Extract a structured claim; a separate verifier checks it is unchanged	Stronger architecture than surface similarity	Presented as a proposal
Human-in-the-loop (4.4)	Flag potential causal-strength changes for accept, reject, or inspect	Preserves authorial responsibility	None stated
Claim verification (4.5)	Check causal verbs against design, analysis, and conclusions	Links wording to evidence	Causality should not be inferred from study design alone
Knowledge editing (4.6)	Modify model knowledge with minimal side effects	Techniques for tracing causal pathways may help later	Changes the model, not user text: related but distinct

5. Research Challenges and Limitations

5.1 Context dependence

Causal meaning is highly context-dependent. “Associated with,” “linked to,” “predicts,” and “drives” do not have universally fixed causal interpretations. A sentence-level classifier can therefore fail when interpretation depends on previous paragraphs, methods, or study design.

5.2 Domain variation

Causal terminology differs across disciplines. Terms such as “determinant,” “risk factor,” “effect modifier,” “driver,” and “mechanism” have specialized meanings in epidemiology, economics, psychology, engineering, and social science.

A benchmark built entirely from biomedical literature may therefore overfit to one disciplinary discourse.

5.3 Causal inference is not reducible to lexical choice

Changing “associated with” to “caused” is clearly consequential, but the reverse problem is equally important: appropriate causal interpretation cannot be established by lexical analysis alone. Causal inference depends on assumptions, identification strategies, design, measurement, confounding, selection, and counterfactual interpretation (Pearl, 2009).

Thus, causal-language preservation should not become an automated system for deciding whether authors *deserve* to make causal claims. Its primary task should be to preserve the author's explicitly supported claim unless substantive revision is requested.

5.4 Ambiguous hedging

Scientific writing frequently uses modal expressions such as “may,” “might,” “could,” “suggests,” “is consistent with,” and “is likely.” These expressions encode uncertainty, but their exact epistemic force is context-dependent.

Kim et al. (2023) found that ChatGPT had particular difficulty with conditional causal claims involving hedges. This suggests that preservation benchmarks must explicitly include hedged claims rather than focusing only on clear “cause” versus “association” contrasts.

5.5 Evaluation difficulties

BLEU, ROUGE, BERTScore, semantic similarity, and general factuality metrics are not sufficient for this task. An edited sentence can score highly on semantic similarity while changing causal status.

A suitable evaluation suite should therefore include:

- causal-status accuracy;
- causal-strength distance;
- hedge preservation;

- directionality preservation;
- population/scope preservation;
- evidence-to-claim alignment;
- human judgments by domain experts.

5.6 Model and version dependence

LLM behavior changes rapidly across models and versions. A benchmark demonstrating preservation in one model does not establish preservation generally. This is particularly relevant because recent studies report substantial differences in causal reasoning across models and model sizes (Liu et al., 2026; Saklad et al., 2026).

Table 6 summarizes the challenges and limitations discussed in Sections 5.1 to 5.6.

Table 6. Research challenges and limitations. Challenges for causal-language preservation and their implications for system and benchmark design.

Challenge	Description	Implication
Context dependence (5.1)	Meaning of “associated with,” “linked to,” “predicts,” “drives” varies; interpretation may depend on earlier paragraphs, methods, or design	Sentence-level classifiers can fail
Domain variation (5.2)	Terms like “determinant,” “risk factor,” and “driver” differ by discipline	A biomedical-only benchmark may overfit
Not reducible to lexical choice (5.3)	Causal inference depends on assumptions, identification, design, confounding	The system should preserve the author's supported claim, not judge whether authors deserve causal claims
Ambiguous hedging (5.4)	“may,” “might,” “suggests,” and similar terms have context-dependent force	Benchmarks must include hedged claims
Evaluation difficulties (5.5)	BLEU, ROUGE, BERTScore, similarity, and factuality miss causal status changes	Need causal-status, strength, hedge, direction, scope, and evidence-alignment measures, plus expert judgments
Model and version dependence (5.6)	Behavior changes across models and versions	Results for one model do not generalize

6. Research Gaps and Future Directions

6.1 A dedicated causal-preservation benchmark

There is a clear need for a benchmark specifically designed around *editing*, rather than causal reasoning or causal extraction alone.

Each benchmark item could contain:

1. an original scientific sentence;
2. study-design information;
3. the author's intended causal status;
4. an LLM-edited version;

5. expert annotations of whether causal force was preserved;
6. acceptable and unacceptable alternative edits.

The benchmark should include causal inflation, attenuation, hedge deletion, scope expansion, and direction reversal.

6.2 Counterfactual semantic evaluation

One promising evaluation method is to ask whether the edited sentence implies the same answer to controlled counterfactual questions.

For example:

> **Original:**
> "X was associated with higher Y."
>
> **Counterfactual test:**
> "Does the sentence imply that intervening to increase X would increase Y?"
>
> Expected answer: *No—not necessarily.*

An edited version that produces *Yes* would exhibit causal inflation.

This provides a more principled test than lexical matching.

Figure 5 illustrates this counterfactual test.

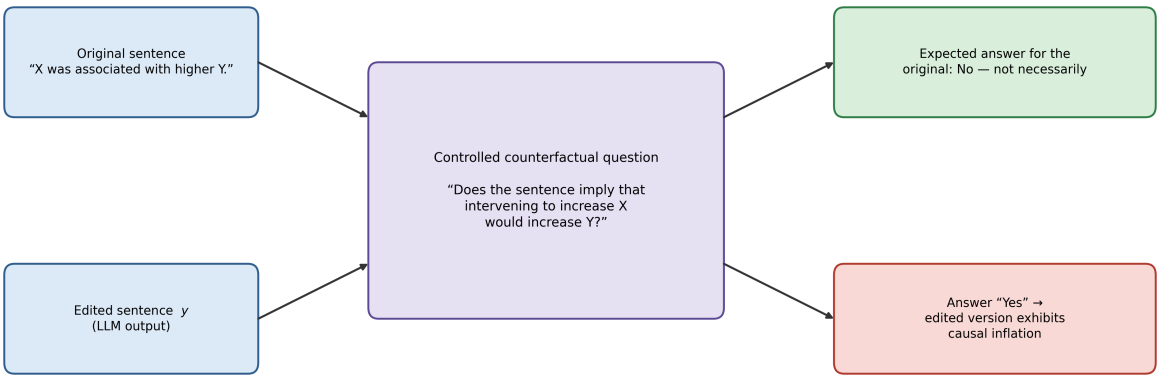


Figure 5. Counterfactual semantic test for causal inflation. The original and edited sentences are probed with the same counterfactual question. A "Yes" for an edit of "X was associated with higher Y" indicates causal inflation.

6.3 Causal graphs as editing constraints

Where sufficient methodological information exists, an editor could represent the study as a directed acyclic graph (DAG). The language generator could then be constrained by the identified causal structure.

For instance, if the text describes an observational association between X and Y confounded by Z, the editing system could be prohibited from generating language implying an unqualified $X \rightarrow Y$ effect.

This connects scientific editing with established causal-graph methodology and may substantially improve semantic control.

6.4 Provenance-aware editing

Future systems should retain a mapping between each edited claim and its source evidence:

$$\text{Edited sentence} \rightarrow \text{Original sentence} \rightarrow \text{Evidence passage} \rightarrow \text{Study design}.$$

Such provenance would enable an editor to explain why a proposed causal verb is potentially problematic.

6.5 Interactive epistemic controls

Rather than providing a single rewritten sentence, an LLM could offer explicitly categorized alternatives:

- **Meaning-preserving:** “was associated with”
- **More cautious:** “was weakly associated with”
- **Potentially causal:** “may have contributed to”
- **Strong causal:** “reduced”

The system should identify which alternatives change the epistemic status of the claim.

6.6 Multilingual causal-language preservation

Most existing resources are concentrated in English. Yet causal semantics are expressed differently across languages, and translation itself can alter evidential and modal distinctions. Future research should construct multilingual causal-editing datasets and examine whether LLMs preserve causal status across translation and subsequent editing.

6.7 Integration into scholarly writing infrastructure

Because LLM-assisted writing is increasingly common in academic publishing, causal-semantic verification could be integrated into word processors, manuscript-management systems, and journal submission platforms. Recent guidance on ethical LLM use emphasizes transparency and human responsibility in academic writing rather than treating LLMs as autonomous authors (Porsdam Mann et al., 2024).

The ideal system would therefore function as a **semantic integrity layer** between author and LLM.

Table 7 summarizes the research directions described in Sections 6.1 to 6.7.

Table 7. Proposed research directions. Future directions identified in the paper.

Direction	Proposed content
Causal-preservation benchmark (6.1)	Items with original sentence, study design, intended causal status, LLM edit, expert annotation, and acceptable and unacceptable alternatives. Should cover inflation, attenuation, hedge deletion, scope expansion, and direction reversal
Counterfactual semantic evaluation (6.2)	Test whether an edit implies the same answers to controlled counterfactual questions
Causal graphs as constraints (6.3)	Use a DAG of the study to constrain generated language
Provenance-aware editing (6.4)	Map each edited sentence to the original sentence, evidence passage, and study design
Interactive epistemic controls (6.5)	Offer categorized alternatives and mark which ones change epistemic status
Multilingual preservation (6.6)	Build multilingual causal-editing datasets
Integration into infrastructure (6.7)	Semantic-integrity layer in word processors, manuscript systems, and submission platforms

7. Conclusion

Preserving causal versus correlational language during LLM-assisted editing is a specialized form of semantic fidelity with direct implications for scientific communication. The distinction between association and causation is foundational to causal inference, and scientific reporting literature demonstrates that humans already struggle to maintain consistent causal framing. LLMs add a new source of risk because their principal strength—fluent, context-sensitive paraphrasing—can inadvertently modify epistemic force.

Existing causal-language classifiers demonstrate that causal status can be computationally identified, while recent studies show that LLMs themselves remain imperfect at distinguishing semantic relatedness from causal explanation. Recent evidence that LLMs can amplify causal overstatement in scientific summaries makes the problem particularly important for automated editing and summarization systems.

The appropriate objective is not to prohibit causal language or to force all observational findings into non-causal formulations. Rather, LLM-assisted editing should preserve the relationship between **evidence, inference, and linguistic strength**. A scientifically responsible editing system should improve clarity while maintaining causal status, uncertainty, scope, directionality, and evidential qualification.

Future systems should therefore move beyond generic notions of factuality and semantic similarity toward explicit causal-semantic constraints, claim-level provenance, structured causal representations, counterfactual evaluation, domain-aware benchmarks, and human-in-the-loop review. The central design principle is straightforward: **an LLM may change how a scientific claim is expressed without silently changing what the evidence permits the claim to mean.**

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